

Problem Set 7

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Ring network

Consider the ring network model we discussed in the class

$$\tau \frac{du_i}{dt} = -u_i + F \left(\sum_{j=1 \dots N} w_{ij} u_j + I_i^0 \right), \quad (1)$$

where $F(x) = [x]_+$ is a rectified linear function and the weights between the neurons are determined by their angular difference and thus are translational invariant.

$$w_{ij} = \frac{1}{N} J(\theta_i - \theta_j), \quad (2)$$

where $\theta_i = -\pi + \frac{2\pi i}{N}$ and J is a 2π periodic function, and I_i^0 is also a periodic 2π function. In the continuous limit, we have

$$\tau \frac{\partial u(\theta, t)}{\partial t} = -u(\theta, t) + \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} J(\theta - \theta') u(\theta', t) d\theta' + I^0(\theta) \right]_+, \quad (3)$$

(a) In the class, we consider a simplified connectivity J

$$J(\theta) = J_0 + J_1 \cos \theta \quad (4)$$

$$I_i^0(\theta) = I_0 + I_1 \cos(\theta - \theta_0) \quad (5)$$

Consider the special case $I_1 = 0$, so that each neuron receives the same homogeneous inputs and $F(x) = [x]_+$. We show in the class that a marginal stable solution is the emergence of a bump, namely, $u(\theta) = [u_1 \cos(\theta) + u_0]_+$. Please show that the bump can actually appear anywhere, namely $u(\theta) = [u_1 \cos(\theta - \phi) + u_0]_+$, where ϕ is arbitrary.

- (b) A naive solution is that the population neural activity is homogeneous : all neurons have the same activity. Show that this naive solution with $u_1 = 0$ is unstable when $J_1 > 2$.
- (c) Draw the phase diagram on the $J_0 - J_1$ plane. Discuss and plot in what region, the system is marginally stable? In what region, the system exhibits homogeneous activity? In what region, the system is unstable?

- (d) Perform computer simulation to confirm your results using a discrete ring network with N neurons for (a)-(c).
- (e) Now consider some tiny modulatory input with $0 < I_1 \ll I_0$ and $J_1 > 2$. Show that in this case, the location of the peak of the stationary activity is not arbitrary, but aligned to the stimulus angle θ_0 .
- (f*) Add to the connectivity matrix a term $\frac{J_1}{N}\gamma \sin(\theta_i - \theta_j)$, where $|\gamma| \ll 1$. Show that there is a solution with the form

$$u(\theta, t) = f(\theta - \omega t), \quad (6)$$

where $f(\theta)$ is the steady state activity profile calculated in the class for $\gamma = 0$ and the angular velocity satisfies

$$\omega = \frac{\gamma}{\tau} \quad (7)$$

Last question is a bonus. Here is some specific guide:

A. Assume a traveling profile of the form of 6 (for now just use an arbitrary profile $f(\theta)$) and insert it into the two sides of 1. Express the RHS in terms of the order parameters as was done in the class. Expand the RHS in powers of γ and keep only terms up to linear in γ .

B. Show that if $f(\theta)$ is the profile for $\gamma = 0$ the dynamic equations 1 are satisfied.